

MWANZA, Tanzania, United Republic ofWMO: **637560**Lat: **2.444S**Lon: **32.933E**Elev: **1147**StdP: **88.28**Time Zone: **3.00 (E03)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	16.9	8.0	7.6	23.3	9.0	8.2	23.0	10.3	24.2	9.9	23.9	2.2	120	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	8.5	30.7	19.4	30.0	19.4	29.2	19.5	22.7	26.6	22.2	26.3	22.0	26.2	5.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.4	24.5	21.1	18.2	24.3	20.8	17.9	24.1	73.2	26.6	71.5	26.4	70.4	26.3	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.9	8.1	DB	12.9	32.3	2.0	0.7	11.5	32.8	10.4	33.2	9.2	33.6	7.8	34.1	(4)
(5)			WB	10.6	24.9	1.2	0.8	9.8	25.5	9.1	25.9	8.4	26.4	7.6	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	23.4	23.7	23.8	23.6	23.6	23.1	22.6	23.2	23.5	23.8	23.2	22.8	(6)	
	DBStd	1.17	1.28	1.32	1.34	1.13	0.86	0.99	0.94	0.90	1.08	1.13	1.13	1.17	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	4874	414	385	427	408	421	392	392	409	406	427	395	398	(10)	
	CDD18.3	1832	156	152	168	158	163	142	134	151	156	168	145	139	(11)	
	CDH23.3	11343	939	971	1068	908	1010	990	979	1053	1013	1029	732	650	(12)	
(13) Wind	CDH26.7	2176	179	218	254	181	174	196	169	207	201	212	103	83	(13)	
	WSAvg	3.4	3.2	3.2	3.2	3.2	3.2	3.5	3.7	3.7	3.9	3.8	3.3	3.0	(14)	
	PrecAvg	1009	91	100	129	179	87	10	2	19	26	78	155	135	(15)	
	PrecMax	1300	184	239	333	329	248	66	27	79	69	185	233	251	(16)	
	PrecMin	674	27	17	14	38	6	0	0	0	0	4	42	49	(17)	
	PrecStd	183	48	57	89	72	57	17	6	24	19	54	53	64	(18)	
	DB	30.5	31.2	31.5	30.4	29.8	30.2	30.2	30.4	30.4	30.9	29.9	29.6		(19)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	19.3	19.9	19.9	20.6	20.0	18.7	17.5	18.2	18.8	19.1	19.9	19.6		(20)	
	DB	29.2	30.1	30.1	29.2	29.0	29.2	29.1	29.4	29.3	29.7	28.9	28.4		(21)	
	MCWB	20.1	20.0	20.3	20.7	20.2	18.6	17.7	18.3	18.9	19.5	20.2	20.3		(22)	
	DB	28.6	29.1	29.1	28.7	28.2	28.8	28.2	28.8	28.8	28.9	28.0	27.7		(23)	
	MCWB	20.3	20.2	20.6	20.8	20.3	18.7	18.0	18.4	18.9	19.6	20.4	20.6		(24)	
	DB	27.8	28.1	28.1	27.8	27.8	27.9	27.6	27.9	27.9	28.0	27.1	26.9		(25)	
	MCWB	20.6	20.7	20.9	20.9	20.3	19.0	18.2	18.7	19.3	20.1	20.6	20.6		(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	22.7	22.9	23.2	23.0	22.2	22.1	21.8	22.2	22.2	22.7	22.4	22.3		(27)	
	MCDB	26.6	26.7	27.1	26.9	26.5	26.4	25.9	25.9	25.9	26.2	26.1	26.1		(28)	
	WB	22.2	22.2	22.5	22.2	21.7	21.4	21.0	21.4	21.7	22.0	21.9	21.8		(29)	
	MCDB	26.3	26.5	26.7	26.5	26.2	25.9	25.2	25.3	25.5	25.8	25.7	25.8		(30)	
	WB	21.7	21.8	22.0	21.9	21.4	20.8	20.4	20.9	21.2	21.6	21.5	21.5		(31)	
	MCDB	26.0	26.2	26.4	26.2	25.9	25.4	24.8	24.9	25.1	25.4	25.4	25.5		(32)	
	WB	21.2	21.4	21.6	21.5	21.0	20.4	19.7	20.4	20.9	21.2	21.2	21.1		(33)	
(34) Mean Daily Temperature Range	MCDB	25.7	25.9	26.0	25.7	25.6	24.9	24.6	24.5	24.8	25.0	25.0	25.1		(34)	
	MDBR	8.4	8.6	8.6	8.2	8.8	10.2	10.9	10.3	9.2	8.5	7.6	7.3		(35)	
	5% DB	10.0	10.5	10.5	9.8	9.7	11.4	11.7	11.6	10.7	10.3	9.3	9.1		(36)	
	MCWBR	4.1	4.0	4.4	4.0	4.1	5.2	5.7	6.1	5.1	4.4	3.5	3.5		(37)	
	MCDBR	8.7	8.8	9.1	8.5	8.8	9.7	10.4	10.2	9.2	8.5	8.0	7.9		(38)	
	MCWBR	4.1	3.9	4.3	4.0	4.0	4.9	5.7	5.6	4.5	4.0	3.5	3.5		(39)	
	taub	0.414	0.426	0.407	0.359	0.357	0.386	0.404	0.416	0.397	0.389	0.376	0.400		(40)	
(41) Clear-Sky Solar Irradiance	taud	2.376	2.343	2.426	2.584	2.567	2.430	2.330	2.305	2.376	2.409	2.488	2.422		(41)	
	Ebn at Noon	920	910	917	939	917	875	864	873	913	932	949	929		(42)	
	Edh at Noon	128	133	122	101	98	110	123	131	126	123	113	121		(43)	
	RadAvg	5.42	5.94	5.84	5.64	5.57	5.63	5.79	5.74	5.73	5.64	5.33	5.25		(44)	
(45) All-Sky Solar Radiation	RadStd	0.53	0.30	0.35	0.44	0.30	0.32	0.26	0.34	0.36	0.39	0.39	0.43		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page